THE CONE ON MULTIPLICATION BY A PRODUCT IS THE CONE OF A MAP BETWEEN THE CONES ON THE FACTORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Cone of a map of complexes and its differential, as recorded in the equation displayed earlier in the chapter. the author cites equation-differential-cone inside the proof at proof lines 6798-6801. Supplies the differential of the cone used in the special case.
- Totalisation of a double complex and its compatibility with homotopies and cones. used at proof lines 6825-6841 to reduce to the special case. Carries the statement from the ring to an arbitrary complex.
- Homotopy equivalence of chain complexes, verified here by exhibiting two maps whose composite is the identity on one side. used at proof lines 6805-6825. Supplies the criterion that the exhibited maps have to satisfy.

Source reference. Original lemma 062A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7789–7789. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0627: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7786–7786. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $A_\bullet$ be a complex of R-modules. Let $f, g \in R$. Let $C(f)_\bullet$ be the cone of $f : A_\bullet \to A_\bullet$. Define similarly $C(g)_\bullet$ and $C(fg)_\bullet$. Then*

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$C(fg)_\bullet$ is homotopy equivalent to the cone of a map

$$C(f)_\bullet[1] \longrightarrow C(g)_\bullet$$

Proof. We first prove this if $A_\bullet$ is the complex consisting of R placed in degree 0. In this case the complex $C(f)_\bullet$ is the complex

$$\ldots \to 0 \to R \xrightarrow{f} R \to 0 \to \ldots$$

with R placed in (homological) degrees 1 and 0. The map of complexes we use is

$$\begin{array}{ccccccccc}
0 & \longrightarrow & 0 & \longrightarrow & R & \xrightarrow{f} & R & \longrightarrow & 0 \\
\downarrow & & \downarrow & & \downarrow{\scriptstyle 1} & & \downarrow & & \downarrow \\
0 & \longrightarrow & R & \xrightarrow{g} & R & \longrightarrow & 0 & \longrightarrow & 0
\end{array}$$

The cone of this is the chain complex consisting of $R^{\oplus 2}$ placed in degrees 1 and 0 and differential (0627)

$$\begin{pmatrix} g & 1 \\ 0 & -f \end{pmatrix} : R^{\oplus 2} \longrightarrow R^{\oplus 2}$$

To see this chain complex is homotopic to $C(fg)_\bullet$, i.e., to $R \xrightarrow{fg} R$, consider the maps of complexes

$$\begin{array}{ccc}
R & \xrightarrow{fg} & R \\
{\scriptstyle (1,-g)}\downarrow & & \downarrow{\scriptstyle (0,1)} \\
R^{\oplus 2} & \longrightarrow & R^{\oplus 2}
\end{array}
\qquad\qquad
\begin{array}{ccc}
R^{\oplus 2} & \longrightarrow & R^{\oplus 2} \\
{\scriptstyle (1,0)}\downarrow & & \downarrow{\scriptstyle (f,1)} \\
R & \xrightarrow{fg} & R
\end{array}$$

with obvious notation. The composition of these two maps in one direction is the identity on $C(fg)_\bullet$, but in the other direction it isn't the identity. We omit writing out the required homotopy.

To see the result holds in general, we use that we have a functor $K_\bullet \mapsto \mathrm{Tot}(A_\bullet \otimes_R K_\bullet)$ on the category of complexes which is compatible with homotopies and cones. Then we write $C(f)_\bullet$ and $C(g)_\bullet$ as the total complex of the double complexes

$$(R \xrightarrow{f} R) \otimes_R A_\bullet \quad \text{and} \quad (R \xrightarrow{g} R) \otimes_R A_\bullet$$

and in this way we deduce the result from the special case discussed above. Some details omitted. □